Intermediate Macroeconomics

Tutorial 4 Unemployment
Answer

1. **The Shapiro-Stiglitz model.** Follow the procedures in the lecture notes page 24-26, derive the value functions for workers in the state S (employed and not exerting effort) and state U (unemployed), i.e., equation (28) and (29).

Answer:
For an employed and shirking worker, he or she receives a net benefit of w from the employment and not being detected and still employed time Δt later with a prabability of $e^{-q\Delta t}e^{-b\Delta t}$. $V_E(\Delta t)$ is therefore

$$V_S(\Delta t) = \int_{t=0}^{\Delta t} e^{-(b+q)t}e^{-\rho t}w dt + e^{-\rho\Delta t}\left[e^{-(b+q)\Delta t}V_S(\Delta t) + \left(1-e^{-(b+q)\Delta t}\right)V_U(\Delta t)\right] \quad (1)$$

$$= \frac{1}{\rho+b+q}\left(1-e^{-(\rho+b)\Delta t}\right)w + e^{-\rho\Delta t}\left[e^{-(b+q)\Delta t}V_E(\Delta t) + \left(1-e^{-(b+q)\Delta t}\right)V_U(\Delta t)\right] \quad (2)$$

Rearrangement leads to

$$V_S(\Delta t) = \frac{1}{\rho+b+q}w + \frac{1}{1-e^{-(\rho+b+q)\Delta t}}e^{-\rho\Delta t}\left(1-e^{-(b+q)\Delta t}\right)V_U(\Delta t) \quad (3)$$

$V_S = \lim_{\Delta t\to 0} V_S(\Delta t)$, applying L'Hôpital rule to above equation, we get

$$V_S = \frac{1}{\rho+b+q}\left[w + (b+q)V_U\right], \quad (4)$$

or

$$\rho V_S = w - (b+q)(V_S - V_U). \quad (5)$$

For an unemployed worker with zero income, the probability to be employed again Δt later is $a\Delta t$. $V_U(\Delta t)$ can be written as

$$V_U(\Delta t) = \int_{t=0}^{\Delta t} 0 dt + e^{-\rho\Delta t}\left[e^{-a\Delta t}V_U(\Delta t) + \left(1-e^{-a\Delta t}\right)V_E(\Delta t)\right] \quad (6)$$

$$= e^{-(a+\rho)\Delta t}V_U(\Delta t) + e^{-\Delta\rho t}\left(1-e^{-a\Delta t}\right)V_E(\Delta t) \quad (7)$$

Rearrangement leads to

$$V_U(\Delta t) = \frac{e^{-\Delta\rho t}\left(1-e^{-a\Delta t}\right)}{1-e^{-(a+\rho)\Delta t}}V_E(\Delta t). \quad (8)$$

$V_U = \lim_{\Delta t\to 0} V_U(\Delta t)$, applying L'Hôpital rule to above equation, we get

$$V_U = \frac{\rho}{a+\rho}V_E, \quad (9)$$

or

$$\rho V_U = a\left(V_E - V_U\right). \tag{10}$$

2. Describe how each of the following affect equilibrium employment and the wage in the **Shapiro–Stiglitz model**:

   (a) An increase in workers' discount rate, ρ.

   Answer:

   The no-shirking condition (NSC) is given by

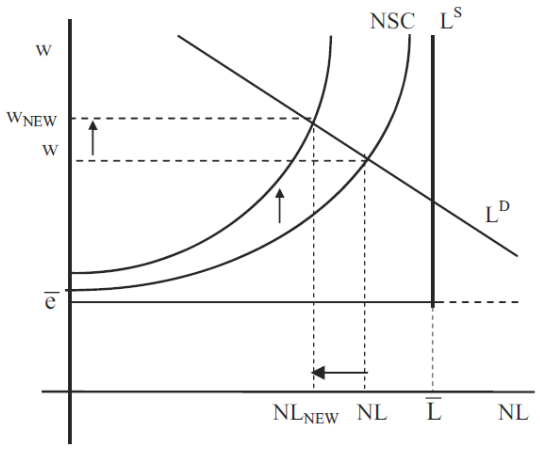
   $$w = \bar{e} + \left(\rho + \frac{\overline{L}}{\overline{L} - NL} b\right) \frac{\bar{e}}{q}. \tag{1}$$

   The labor demand curve is given by

   $$AF'\left(\bar{e}L\right) = w/\bar{e}. \tag{2}$$

   Equation (2) states that firms choose L so that the marginal product of effective labor equals the marginal cost of effective labor, where the wage, w, is set to satisfy (1).

   An increase in ρ shifts up the no-shirking locus. From equation (1), for a given NL, the wage needed to get workers to exert effort is now higher. Intuitively, since workers discount the future more, it matters less to them if they are caught shirking, are fired and have to go through a period of unemployment. Thus at a given level of employment, firms must pay a higher wage to deter shirking. The labor demand curve is unaffected. As shown in the figure below, equilibrium employment falls and the equilibrium wage rises.



   (b) An increase in the job breakup rate, b.

   Answer:

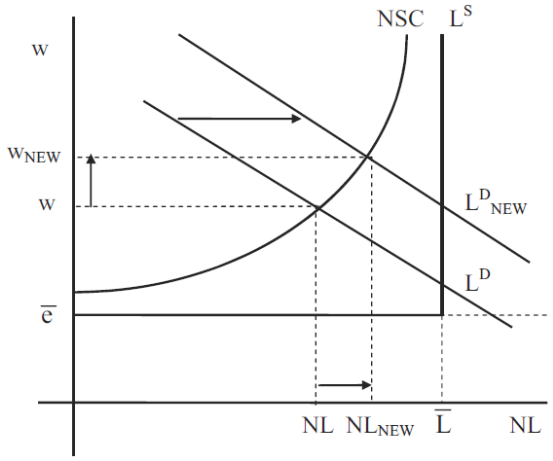
   An increase in the job breakup rate, b, shifts up the no-shirking locus. From equation (1), for a given NL, the wage required to get workers to exert effort is now higher. Intuitively, since workers are more likely to lose their job anyway, the value of being employed is lower. Thus workers are not as concerned about being caught shirking and fired. So at a given level of employment, firms must pay a higher wage to deter

shirking. The labor demand curve is unaffected. Equilibrium employment falls and the wage rises. [Same figure as part (a).]

(c) A positive multiplicative shock to the production function (that is, suppose the production function is $AF(L)$, and consider an increase in A).

Answer:

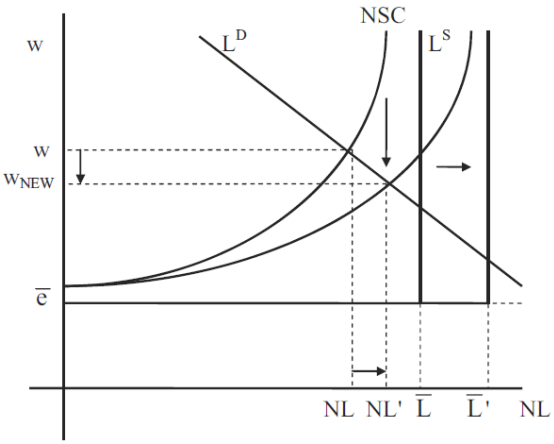
The rise in A shifts the labor demand curve to the right. The no-shirking locus is unaffected. As shown in the figure below, the equilibrium wage rises as does the level of employment. Note that if efficiency wages were not present, inelastic labor supply would mean that increases in technology would lead only to increases in the wage, not to increases in employment.



(d) An increase in the size of the labor force, $\overline{L}$.

Answer:

The vertical portion of the labor supply curve shifts to the right. The labor demand curve is unaffected. The no-shirking locus shifts down. Intuitively, at a given NL, $\overline{L} - NL$ is now higher. Thus at a given level of employment, if workers become unemployed, they are likely to stay unemployed longer. Thus at a given level of employment, the cost of shirking is greater for a worker and thus firms can get away with paying a lower wage to deter shirking. From the figure at right, the equilibrium wage falls and employment rises.



3. **Union wage premiums and efficiency wages.** (Summers, 1988.) Consider the efficiency-wage model analyzed in equations (12)–(17) of the lecture notes. Suppose, however, that fraction f of workers belong to unions that are able to obtain a wage that exceeds the nonunion wage by proportion μ. Thus, $w_u = (1+\mu)w_n$, where w_u and w_n denote wages in the union and nonunion sectors; and the average wage, w_a, is given by $fw_u + (1-f)\,w_n$. Nonunion employers continue to set their wages freely; thus (by the same reasoning used to derive (15) in the text), $w_n = (1-bu)\,w_a/\,(1-\beta)$.

   (a) Find the equilibrium unemployment rate in terms of β, b, f and μ.

   Answer:

   Substituting the expression for the average wage, $w_a = fw_u + (1-f)\,w_n$, into the expression for the nonunion wage, $w_n = (1-bu)\,w_a/\,(1-\beta)$, yields

$$w_n = \frac{1-bu}{1-\beta}\left[fw_u + (1-f)\,w_n\right]. \tag{1}$$

   Substituting the union wage, $w_u = (1+\mu)w_n$, into equation (1) yields

$$\begin{aligned} w_n &= \frac{1-bu}{1-\beta}\left[f(1+\mu)w_n + (1-f)\,w_n\right] \\ &= \frac{1-bu}{1-\beta}\,(1+\mu f)\,w_n. \end{aligned} \tag{2}$$

   Simplification gives us

$$(1-bu)\,(1+\mu f) = 1-\beta. \tag{3}$$

   Working out the algebra, we have

$$u = \frac{\beta+\mu f}{b\,(1+\mu f)}. \tag{4}$$

   (b) Suppose $\mu = f = 0.15$. What is the equilibrium unemployment rate if $\beta = 0.06$ and $b = 1$? By what proportion is the cost of effective labor higher in the union sector than in the nonunion sector?

   Answer:

   Substitute the numeric values of β, b, f and μ into equation (4), we have

$$u = \frac{0.06 + (0.15)\,(0.15)}{1 + (0.15)\,(0.15)} \approx 0.081. \tag{5}$$

   Equilibrium unemployment is approximately 8.1 percent, which is higher than the 6 percent obtained with $\beta = 0.06$ and $b = 1$ in the standard version of this model without a union sector.

   In order to determine the proportion by which the cost of effective labor in the union sector exceeds that in the nonunion sector, we need to calculate the equilibrium effort level in each sector. The union wage as a function of the average wage is

$$w_u = (1+\mu)w_n = (1+\mu)\,(1-bu)\,w_a\,/(1-\beta)\,. \tag{6}$$

To find the effort for union workers, substituting equation (6) and the definition of the index of labor-market conditions, $x = (1 - bu)w_a$, into the expression for effort, $e = [(w - x)/x]^{\beta}$, gives us

$$\begin{aligned} e_u &= \left[\frac{(1+\mu)\,(1-bu)\,w_a\,/(1-\beta)\,-(1-bu)w_a}{(1-bu)w_a}\right]^{\beta} \\ &= \left(\frac{\mu+\beta}{1-\beta}\right)^{\beta}. \end{aligned} \tag{7}$$

To find the effort for nonunion workers, substituting $w_n = (1 - bu)\,w_a/\,(1 - \beta)$ into the expression for effort yields

$$\begin{aligned} e_n &= \left[\frac{(1-bu)\,w_a\,/(1-\beta)\,-(1-bu)w_a}{(1-bu)w_a}\right]^{\beta} \\ &= \left(\frac{\beta}{1-\beta}\right)^{\beta}. \end{aligned} \tag{8}$$

In the union sector, the cost of effective labor is

$$\frac{w_u}{e_u} = \frac{(1+\mu)w_n}{\left[(\mu+\beta)\,/(1-\beta)\right]^{\beta}}. \tag{9}$$

Similarly, the cost of effective labor in the nonunion sector is

$$\frac{w_n}{e_n} = \frac{w_n}{\left[\beta\,/(1-\beta)\right]^{\beta}}. \tag{10}$$

Dividing equation (9) by equation (10) gives us the following ratio of the cost of effective labor in the union to the nonunion sector:

$$\frac{w_u\,/e_u}{w_n\,/e_n} = \frac{(1+\mu)w_n}{\left[(\mu+\beta)\,/(1-\beta)\right]^{\beta}}\frac{\left[\beta\,/(1-\beta)\right]^{\beta}}{w_n} = (1+\mu)\left(\frac{\beta}{\mu+\beta}\right)^{\beta}. \tag{11}$$

Substituting $\mu = 0.15$ and $\beta = 0.06$ into equation (11) gives us

$$\frac{w_u\,/e_u}{w_n\,/e_n} = 1.15\left(\frac{0.06}{0.15+0.06}\right)^{0.06} \approx 1.0667. \tag{12}$$

Note that although the cost of labor in the union sector exceeds the cost of labor in the nonunion sector by a factor of $(1 + \mu) = 1.15$, the cost of effective labor is only higher by a factor of about 1.07. This is because union workers exert more effort since they are paid a higher wage.

(c) Repeat part (b) for the case of $\beta = 0.03$ and $b = 0.5$.

Answer:

Notice that the cost of effective labor is not influenced by the parameter b. Substi-

tuting $\beta = 0.06$ into equation (11) gives us

$$\frac{w_u / e_u}{w_n / e_n} = 1.15 \left(\frac{0.03}{0.15 + 0.03} \right)^{0.06} \approx 1.0898. \tag{13}$$

With the elasticity of effort with respect to the wage lower at $\beta = 0.06$, the ratio of the cost of effective labor in the union sector to that in the nonunion sector is now higher.